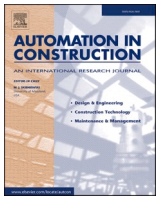
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Personalized construction safety training system using conversational AI in virtual reality

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ABSTRACT

Training workers in safety protocols is crucial for mitigating job site hazards, yet traditional methods often fall short. This paper explores integrating virtual reality (VR) and large language models (LLMs) into iSafeTrainer, an AI-powered safety training system. The system allows trainees to engage with trade-specific content tailored to their expertise level in a third-person perspective in a non-immersive desktop virtual environment, eliminating the need for head-mounted displays. An experimental study evaluated the system through qualitative, survey-based assessments, focusing on user satisfaction, experience, engagement, guidance, and confidence. Results showed high satisfaction rates (>85 %) among novice users, with improved safety knowledge. Expert users suggested advanced scenarios, highlighting the system's potential for expansion. The modular architecture supports customization across various construction settings, ensuring adaptability for future improvements.

1. Introduction

Construction sites pose considerable safety risks due to their dynamic nature and diverse workforce expertise levels [1]. These risks are particularly acute for novice workers, whose limited safety knowledge and experience necessitate comprehensive training programs [2]. While training constitutes a crucial component of workplace safety protocols, other environmental factors and critical interventions, such as personal protective equipment (PPE), administrative controls, and engineering controls, play a vital role in mitigating job-site hazards. While these measures provide essential protection, educating workers about safety hazards and their mitigation methods remains a fundamental requirement. Despite the importance of such comprehensive approaches, limited budgets often restrict access to robust safety training programs, leading companies to rely on standardized methods, such as lectures or video-based instruction, which fail to address individual learning needs. These constraints are especially problematic for high-risk trades, where targeted, hands-on safety training is essential [3]. Consequently, unskilled labor, driven by low profit margins, can cause safety incidents, emphasizing the need for proper safety training [3,4]. Traditional lecture-based safety training, while widely used, has proven insufficient for addressing on-site challenges, as it primarily relies on static materials like text, images, and videos that lack adaptability to real-world, high-risk scenarios [5–8].

Emerging VR and AI technologies enable immersive, personalized learning experiences [9–12], [12], [13,14], with Non-Playing Characters (NPCs) providing tailored instruction based on individual field demands [15–18]. Further integrating VR with AI offers a cutting-edge approach to personalized training, showing promise in effectiveness [19,20]. The immersive nature of VR offers immediate feedback, making it an effective alternative to real-world training in hazardous scenarios. However, traditional computer inputs may limit the effectiveness of virtual reality for less skilled or literate workers [21]. AI, particularly Large Language Models (LLMs) like GPT-4, can address this limitation by creating conversational agents that enhance engagement and understanding while reducing cognitive load [19,22,23].

This study introduces iSafeTrainer, an LLM-integrated VR system that overcomes the limitations of pre-scripted scenarios through immediate, conversational, and adaptive training. The system employs a third-person perspective in a non-immersive desktop virtual environment, viewed via a standard monitor or desktop to simulate construction safety training environments, reducing the risk of motion sickness and dizziness often associated with head-mounted displays (HMDs) [24] while maintaining engagement through interactive, trade-specific scenarios. Unlike conventional methods that require manual updates for new scenarios, the proposed system dynamically personalizes training

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0926-5805/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

based on users' trade and knowledge level, ensuring content remains relevant and tailored to individual needs. Students, defined as those enrolled in engineering programs, and workers, including both newcomers and experienced professionals on construction jobsites, receive targeted instruction aligned with their specific expertise. The system customizes multiple factors, such as conversation style, instruction pacing, content complexity, and trade-specific terminology, to ensure relevance.

Trade and knowledge level personalization ensures more effective safety training. The construction sector encompasses a wide range of specialized trades, each with unique safety considerations. Recent studies highlight a significant incidence of trade-specific injuries in it, particularly among high-risk activities such as welding and scaffolding [25–28]. By tailoring training based on trade, iSafeTrainer provides highly relevant and specific safety information—electricians receive focused training on electrical hazards, while scaffolders learn about fall protection specific to their work. Additionally, knowledge level personalization addresses the diverse backgrounds of learners, ensuring that novices receive step-by-step guidance, beginners reinforce foundational knowledge, and experts engage in advanced problem-solving tasks. This approach prevents training from being either overwhelming or redundant, improving engagement and retention. By tailoring instructional style, pacing, and trade-specific content, iSafeTrainer ensures training remains relevant and appropriately challenging, thereby enhancing engagement and retention. Post-training surveys assess user satisfaction, engagement, and perceive improvements in safety knowledge, providing insights into the system's effectiveness. In contrast to previous research that relied on static, manually designed scenarios, the proposed system dynamically personalizes training using on-the-spot user data, enabling real-time adaptation to individual users and industry requirements.

This study focuses on high-risk trades such as electrical work, scaffolding, and welding, which contribute significantly to workplace injuries and fatalities, as highlighted by the Occupational Safety and Health Administration (OSHA) [29]. Targeting these areas ensures that critical safety concerns in the construction are addressed. The system is evaluated through questionnaire-based surveys, including structured and unstructured questions, assessing user satisfaction, engagement, and knowledge retention [30,31]. The key contributions of the paper include:

- Development of an LLM-integrated VR system that delivers personalized, trade-specific construction safety training.
- Implementation of adaptive content delivery based on user expertise and trade requirements
- Evaluation of the feasibility and user satisfaction of the iSafeTrainer system using questionnaire-based survey methods.

The remainder of this paper discusses the challenges of traditional safety training and solutions proposed in the literature. It then details the research methodology, system architecture, and implementation, followed by an experimental case study on personalized safety training. The results are discussed and interpreted, and the paper concludes by summarizing contributions, limitations, and future work.

2. Literature Review

2.1. Construction safety training

Construction Safety Training: The inherent hazards in construction necessitate robust safety training protocols [1]. As documented by Mohammed et al. and Boateng et al., conventional training approaches utilize lectures, toolbox talks, demonstrations, and various printed materials like manuals, posters, and videos [5,8], [32,33]. While Bęś et al. point out that these methods are familiar and easy to implement, they can be time-consuming and may disrupt workflow [34]. Furthermore, these materials might not address specific job hazards or individual learning preferences, limiting their effectiveness. A critique by Rafindadi et al. emphasizes a major drawback of traditional methods, their limited ability to engage learners actively and accurately reflects the unpredictable nature of construction sites [35]. This gap between theoretical knowledge and practical application can leave workers, particularly new ones, unprepared for real-world scenarios, as Åsgård et al. observed [36]. Liu et al. also noted that construction workers often struggle to grasp standardized safety concepts in classroom settings due to limited experience, which compounds the challenge of applying theoretical knowledge in practical situations where quick decision-making is crucial [37]. Kaplan et al. and Hussain et al. further highlight this issue [38,39].

2.2. Advanced technologies for enhancing construction safety

Recent advancements in visualization technologies, such as VR, AR, and BIM, have improved construction safety training by creating immersive hazard simulations [38,40]. Akinlolu and Haupt, along with Afzal et al., highlight how these technologies bridge the gap between theoretical learning and hands-on practice, improving safety outcomes [41,42]. Zhu et al. further emphasized their role in risk-free hazard simulation, allowing workers to experience dangerous scenarios in controlled settings [43]. However, Li et al. and Wen et al. noted that current VR-based training often relies on pre-scripted tasks and one-size-fits-all content, which fail to accommodate diverse learning preferences and real-world variability [14,44]. Furthermore, these approaches lack adaptability to different trade specializations and fail to provide immediate feedback tailored to users' expertise levels. Studies such as those by Xu et al. and Lee et al. have emphasized the need for dynamic, scenario-based training models that can personalize learning experiences based on a participant's trade, skill level, and progress [45,46]. To address these challenges, researchers have attempted to integrate BIM-based visualization and simulation software to create more realistic training scenarios [41,47]. However, even these advanced visualization techniques remain constrained by pre-planned tasks that do not offer dynamic adaptability or user-specific adjustments [14]. Therefore, it is crucial to develop personalized and adaptive training programs with instantaneous guidance and guidance, which should be capable of dynamically adjusting to the progress and requirements of individual learners, thereby enhancing the effectiveness of safety training. In response, researchers have integrated LLMs into VR for adaptive learning [48–50]. Several survey studies explore the use of LLM-integrated agents in training environments and highlight the associated limitations [51–54]. AI-driven non-player characters (NPCs) in VR dynamically adjust instructional content and feedback, enhancing interactivity and training personalization [48,55,56].

Immersive learning, when combined with personalized training, enhances engagement, comprehension, and retention. Studies by Xu et al. and Lee et al. suggest that tailoring VR training to individual user needs creates highly effective learning experiences by reinforcing critical safety skills through repeated exposure to hazardous situations in a risk-free environment [45,46]. Unlike static VR simulations, which rely on fixed training modules, iSafeTrainer introduces a fully adaptive AI-driven NPC that adjusts training complexity in real time, based on the user's trade, experience level, and performance. Prior research has highlighted the need for greater user engagement and adaptability in immersive training environments. However, fully immersive systems that rely on head-mounted displays (HMDs) introduce discomfort, such as motion sickness or dizziness, particularly during extended training sessions [24,50,51,57]. To overcome these challenges, the proposed research employs a third-person perspective in a non-immersive desktop setup, allowing for trade-specific, interactive learning while maintaining user comfort and engagement.

Such advancements indicate a shift towards tailored, communication-focused safety training that aligns with real-world

construction site demands. However, existing VR-based methods continue to face significant limitations. Studies by Xu & Zheng describe a multi-level VR training system but highlight persistent challenges in hardware integration and dynamic site simulations, restricting large-scale adoption. [58]. Similarly, Cruz et al. and Eiris et al. examined the gap between simulation-based learning and real-world application, emphasizing the need for more adaptable, interactive training methods [57,59]. Researchers such as Joshi et al. and Kaarlela et al. further explored VR adoption in various construction sectors, noting issues related to technology dependence and simulation accuracy [60,61]. Li et al. focus on synthesizing personalized VR training scenarios, confronting challenges like high dependency on advanced VR technology [14]. Wen et al. discuss virtual humans in VR, which show potential in enhancing communication skills but struggle with replicating complex human interactions [44]. Similarly, Hussain et al. explored AI-based VR systems for migrant worker training, addressing language barriers but pointing out complexities in programming AI for diverse linguistic and cultural backgrounds [19].

To address these persistent challenges, iSafeTrainer introduces an AI-driven NPC framework that personalizes training scenarios in real time, adapts to users' trade-specific needs, and enhances engagement through dynamic hazard recognition and immediate feedback. Contrasting static VR methods, iSafeTrainer dynamically evolves with user interactions. Below, Table 1 summarizes key research studies on integrating VR, AI, and LLM-integrated NPCs for safety training. This comparison highlights the major limitations of existing methods, including dependency on static content, lack of real-world application data, and challenges in simulating complex interactions, which iSafeTrainer aims to overcome.

While previous research has made significant strides in integrating VR, AI, and LLM-integrated agents for safety training, these studies still face key challenges, such as high dependency on advanced VR technology, limited adaptability to dynamic real-world construction sites, and difficulties in replicating complex human interactions. The intended system mitigates these limitations by offering a flexible, trade-specific, and user-adaptive training framework. Unlike static, pre-scripted VR scenarios, iSafeTrainer dynamically adjusts training content based on the user's trade, experience level, and real-time interactions. Additionally, by utilizing conversational AI-driven NPCs, the system provides individualized, context-aware feedback to enhance learning engagement. To address accessibility challenges, iSafeTrainer employs a third-person perspective in a non-immersive desktop VR interface rather than HMDs, reducing discomfort while maintaining immersion. Furthermore, the system is designed to be scalable and adaptable, allowing for future enhancements, including multi-language support and trade-specific AI-driven scenario generation. These improvements position iSafeTrainer as a more effective and adaptable alternative to traditional and AI-assisted VR training models.

2.3. Correlation between constructivism and iSafeTrainer

The concepts of iSafeTrainer align with constructivist learning theories, which emphasize active participation, guided discovery, and reflective learning [63]. Unlike passive safety training approaches, constructivist learning enables users to actively engage with their environment, apply safety measures, and refine decision-making skills through dynamic feedback. The AI NPC from the developed system integrates these principles by requiring participants to identify hazards, apply safety measures, and adapt responses based on real-time AI interactions. Rooted in constructivist learning theory, iSafeTrainer integrates active participation, guided discovery, and reflective learning to create an immersive VR training environment [16]. Different from traditional methods, constructivist learning encourages interaction with the environment, problem-solving, and immediate feedback. The AI NPC facilitates this approach by requiring participants to identify hazards, make safety decisions, and apply mitigation strategies in a dynamic, responsive setting. Through context-aware guidance, the system adapts training content based on user inputs, enhancing decision-making and problem-solving skills. The instant guidance mechanisms further reinforce the learning process, helping users correct errors and deepen their understanding of safety procedures. Constructivist approaches, as suggested by Wenjuan et al. and Bęś et al. [34,63,64], foster critical thinking and engagement, essential for safety hazard recognition and risk perception. The model promotes learner-centered education, enhancing real-world safety skills through interactive, personalized training scenarios. Fig. 1 Illustrates this educational process, where the learning environment fosters exploration, on-the-spot assessment, and interactive hazard simulations. The AI-driven guidance ensures that learning is dynamic, reflective, and personalized, making it more

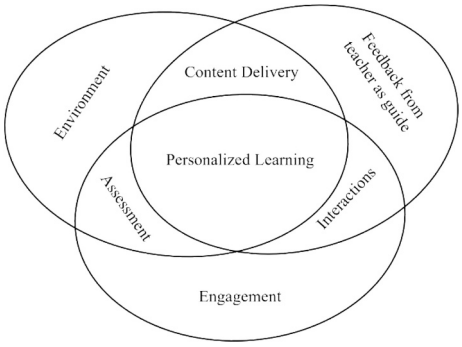


Fig. 1. Constructivist learning principles with respect to personalized safety training.

Table 1
Summary of key research studies integrating VR, AI, and LLM-integrated NPs.

Year	Author	Objective	Technology	Approach	Key Limitation	Reference
2022	Li et al.	Personalized VR training scenarios	VR	Developed Virtual environments requiring advanced VR technology	High dependency on expensive and advanced VR setups	[14]
2023	Y. Ge, et al.	AIOS agent ecosystem	LLM-integrated AI Agents	Utilized pre-trained systems to streamline agent interactions	Lack of adaptability in dynamic real-world construction sites	[62]
2023	M. Klissarov, et al.	AI feedback and motivation	AI, LLM	Integrated LLMs with AI to provide feedback while addressing context gaps	Challenges in real-time user adaptation and feedback	[50]
2023	Wen et al.	Communication training using virtual humans	VR	Simulated human interactions using VR-based communication techniques	Limited ability to replicate complex human interactions	[44]
2024	S. Hu, et al.	LLM-integrated game agents' conceptual architecture	LLM-integrated game agents	Proposed architecture for integrating LLMs with game agents	Difficulty in balancing AI autonomy and user control	[52]
2024	Q. Ma, et al.	Autonomous LLM-powered multi-agent architectures	Autonomous LLM-integrated agents	Explored alignment mechanisms for balancing agent autonomy	Lack of validation in construction safety applications	[53]
2024	F. Li, et al.	LLMs for Behavior Tree Generation	LLM-integrated agents	Used LLMs to automate behavior tree generation, focusing on fine-tuning	Challenges in optimizing LLMs for domain-specific scenarios	[54]
2024	Hussain et al.	AI-based VR training for migrant workers	AI and VR	Designed prompt engineering tailored for diverse linguistic backgrounds	Complexes in programming AI for multilingual and cultural diversity	[19]

effective for long-term skill development and on-site application [65]

3. Research methodology

This section outlines the research methodology for the proposed VR safety training system. Fig. 2 provides a detailed visualization of the iSafeTrainer system architecture, outlining the workflow and interactions among its core components. After the system development, the user training begins with trade and knowledge selection, followed by prompt engineering for content generation. The AI-driven system dynamically adapts content based on user input, enhancing the training's relevance. Finally, practical implementations and content delivery are evaluated through case studies, ensuring the system's reliability and applicability to various scenarios. The modular and layered design of the system ensures scalability and adaptability.

3.1. System architecture and development

The study employed an experimental approach based on constructivist learning theory to address the personalized safety training needs of students and workers. The system applies constructivist learning, encouraging active participation, guided discovery, and reflection. Users interact directly with the Virtual environment, making safety decisions rather than passively receiving information. The AI NPC supports guided discovery by responding dynamically to user queries and providing real-time hazard identification guidance. Immediate AI-driven reflective learning allows users to correct errors and adjust their approach, reinforcing knowledge retention and critical thinking. To align with this, the iSafeTrainer, an LLM-integrated virtual instructor, was integrated into a Virtual environment, providing tailored learning experience. Fig. 3 shows how participants engage with the Virtual environment, performing tasks and interacting with iSafeTrainer to enhance their learning.

3.1.1. iSafeTrainer development

The system was developed on Unity 3D (Windows 11, Intel Core i9 10th-gen 3.70 GHz, 16GB RAM). Unity Editor version 2021.3.16f1 was used to develop the iSafeTrainer system. The system featured a 3D mesh from a real construction site, created with Gaussian Splatting and supplemented with a Sketchfab mesh, and custom virtual objects for various construction hazards. These 3D objects, including buildings and equipment, were sourced from the Unity Asset Store or created in Blender 3.5. [66,67]. Realistic sounds of construction machinery and site noises were also simulated, replicating the auditory atmosphere of a real construction site.

The interface of the training system features a main menu where users selects their trade (Electrical, Scaffolding, or Welding) and knowledge level (Novice, Beginner, Expert), this selection was self-identified, as shown in Fig. 4. Users can start the training by selecting 'Start' or exit using the 'Quit' button. Based on their selections, they are guided to specific environments where hazards are marked with yellow arrows. iSafeTrainer is available for interaction, and hazards are highlighted to ensure that users can identify and discuss them. If any hazards are missing, the AI NPC prompts users to revisit the site and identify the remaining hazards, thereby assessing their knowledge and active learning.

The proposed research study developed VR-based training environments for electrical, welding, and scaffolding sites, providing realistic settings where users can identify various hazards. The third-person perspective in a non-immersive VR desktop setup encourages

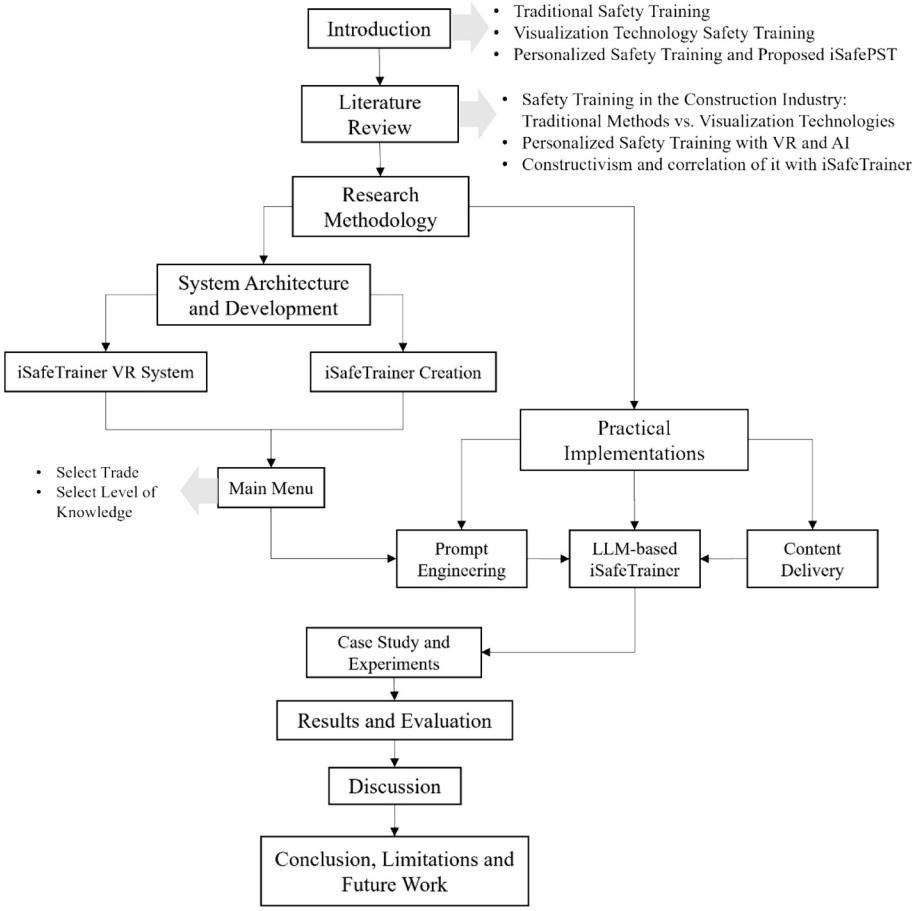


Fig. 2. Research methodology framework for the iSafeTrainer based VR system.

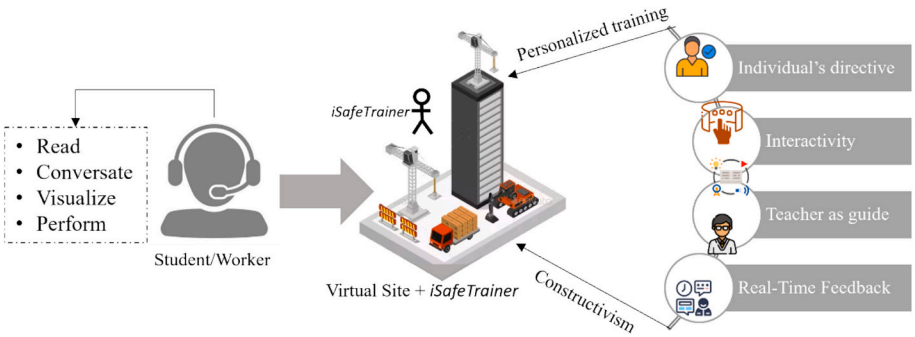


Fig. 3. Correlation of constructivism and personalized safety training.

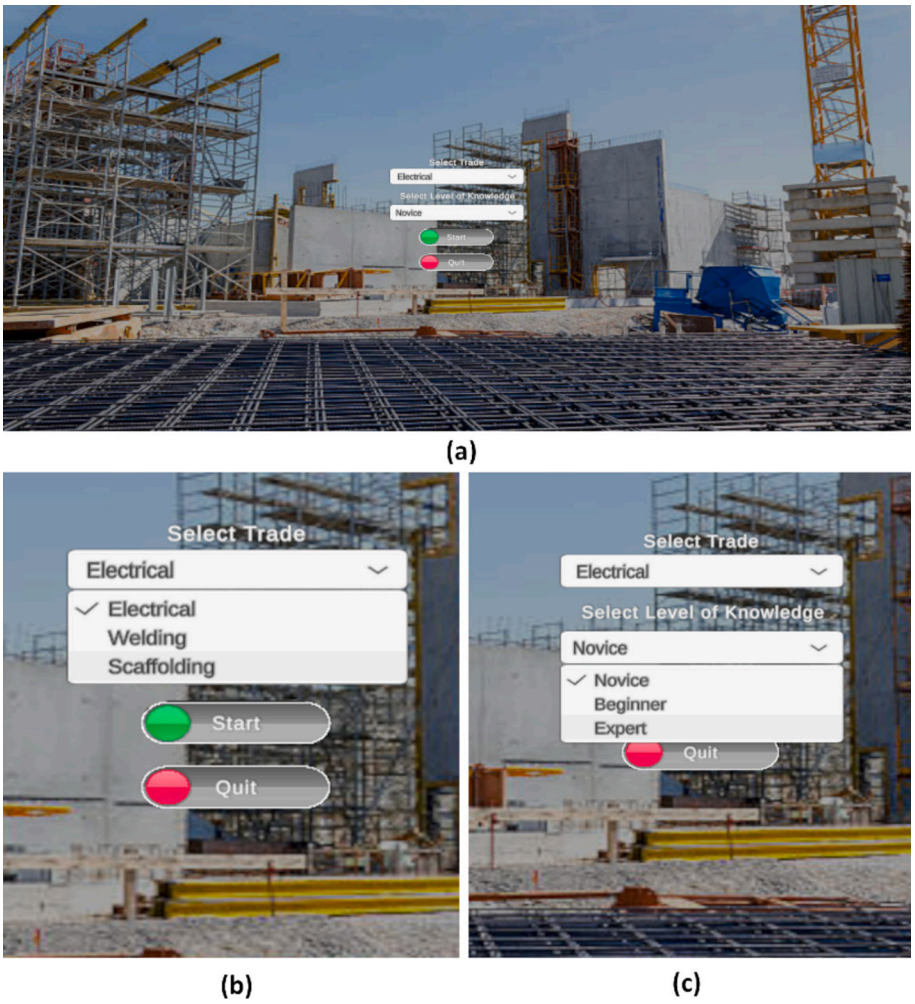


Fig. 4. **(a)** Main Menu Structure: **(b)** Sub-Menu for Trade Selection; **(c)** Sub-Menu for Level of Knowledge Selection.

participants to explore the site, engaging in observational and critical thinking activities to understand the impact of hazards on worker safety through discussions with AI NPC. For electrical safety training, key hazards include wet floors, misplaced overhead power lines, and exposed wires. Additionally, safety cones, fire extinguishers, and distribution boxes are placed to challenge players to distinguish between hazards and mitigation equipment (Fig. 5(a)). The VR scaffolding site highlights hazards such as missing guardrails, fall hazards, and open hatches, ensuring users are prepared to identify safety issues in scaffolding contexts (Fig. 5 (b)). In the welding site, hazards include fire blasts due to sparks on oil cans, electrical shocks, physical hazards, and the lack of proper protective equipment (PPEs) (Figure 5(c)).

In each of these virtual training environments, AI trainer (NPC) assists participants based on their selected trade and level of knowledge. If participants correctly identify all hazards, AI NPC validates their success; if not, it prompts them to fully identify and mitigate hazards. Additionally, AI NPC can suggest video-based training content tailored to the participant's specific needs. These videos are integrated within the Unity-based Virtual environment and can be played or paused by the user. This feature enhances engagement and learning by providing tailored educational content, as depicted in Fig. 6.

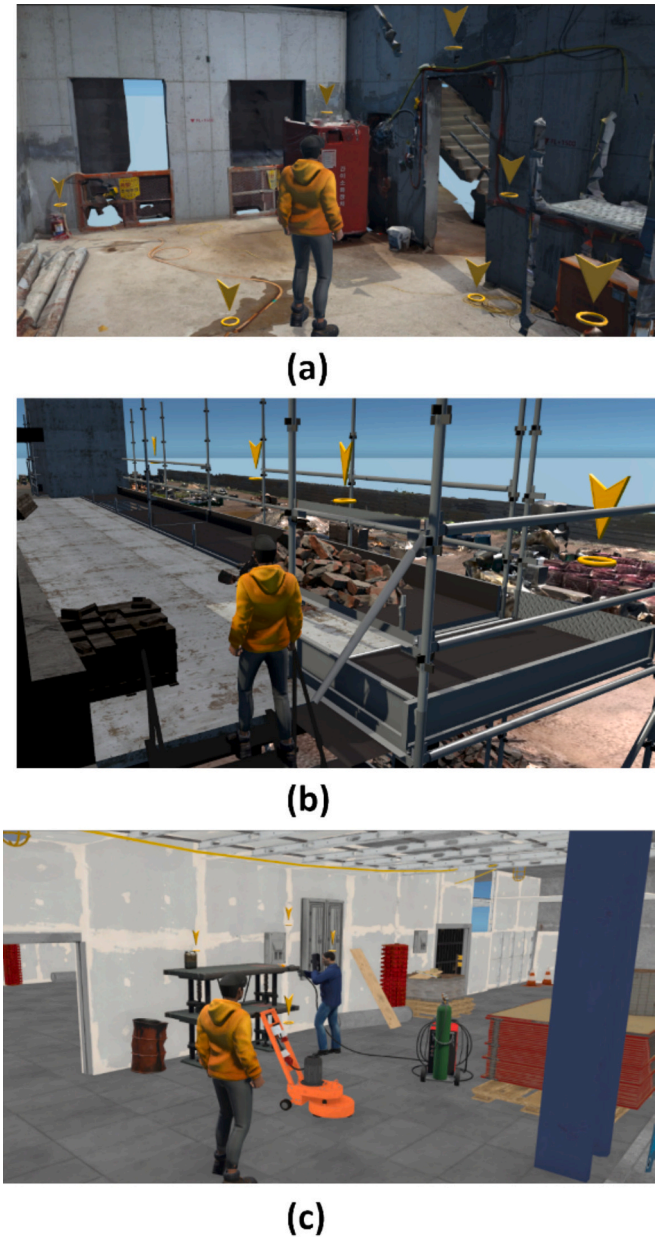


Fig. 5. Virtual construction sites **(a)** Electrical Safety Training platform, **(b)** Scaffolding Safety Training Platform, and **(c)** Welding Safety Training Platform.

3.1.2. LLM-integrated iSafeTrainer avatar creation

The primary objective of the proposed VR system is to incorporate an LLM-integrated virtual instructor, known as iSafeTrainer, to enhance the training experience for students and workers for interactive and personalized communication to deliver training content in a way that aligns with constructivist learning theory, which emphasizes learning through experience and interaction with the environment [64]. The AI NPC in iSafeTrainer is developed using the "Ready Player Me" platform, a comprehensive library of customizable virtual characters, to create an instructor role tailored to meet the specific communication needs of students and workers which enables more effective acquisition of safety knowledge [68]. The instructor is animated within the Unity gaming environment, utilizing humanoid animations such as idle and talk motions from the publicly accessible Mixamo library [69]. To create a more realistic human-like interaction, an animator component is integrated into iSafeTrainer, allowing for synchronized hand and leg movements during communication. Additional scripts control eye and head movements, enabling the avatar to blink and direct its gaze towards the student or worker (player) during dialogue.

OVR (Oculus VR (Virtual Reality)) lip synchronization from Oculus is used to simulate realistic mouth movements when the AI trainer speaks [70]. The overall process of creating the AI NPC avatar and adding human-like characteristics through animation is illustrated in Fig. 7.

3.2. Practical implementations

Existing construction safety VR training programs primarily deliver pre-defined courses and tasks using generalized AI NPCs for communication-based training aimed at students and workers. However, a more proactive approach is personalized safety training, which better prepares individuals for real-world construction environments [14]. The proposed system dynamically adjusts training complexity, pacing, and terminology through AI-driven NPC interactions. The system continuously refines its responses based on real-time user input, allowing for a context-aware and adaptive learning experience that mimics human instruction. By following a linear progression of pre-set scenarios, iSafeTrainer leverages LLMs to interpret user queries, video content recommendation and generate customized training responses in real time.

3.2.1. Prompt engineering for iSafeTrainer

The iSafeTrainer uses prompt-based training to tailor communication based on the user's trade and knowledge level (novice, beginner, expert), identified through the 'PlayerInfo' script. The system provides relevant responses to hazard inquiries in the environment, guided by the 'worldInfo' script. Prompt engineering, a key strategy for guiding AI models like ChatGPT [71], ensures that the AI generates contextually appropriate responses for construction students and workers [72]. Unity, C# is used to implement prompts, considering the player's trade,



Fig. 6. (a) Conversational View between iSafeTrainer and User **(b)** Animated Video is Playing in the Scene to Instruct the User How to Work on Welding Worksite.

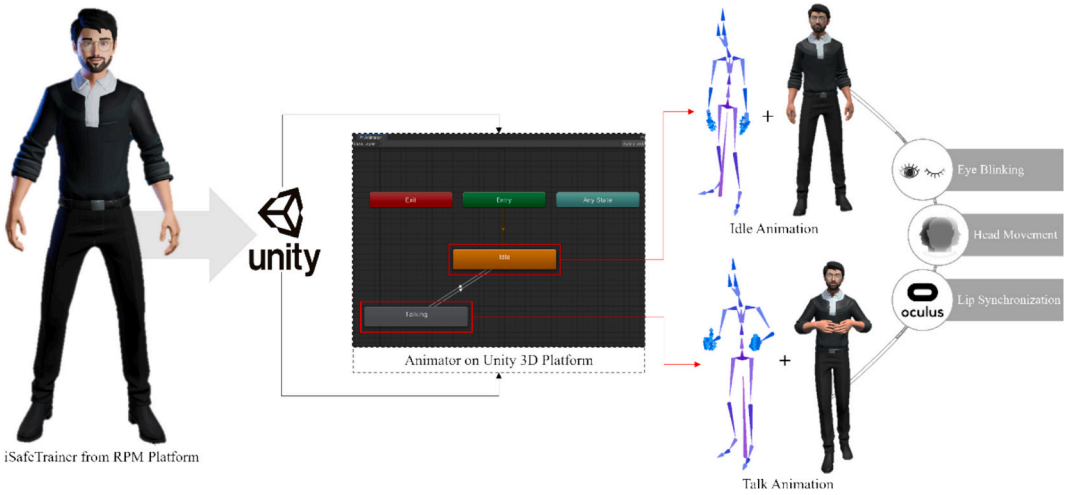


Fig. 7. Animated iSafeTrainer Creation Procedure.

knowledge level, and environment, creating an educational experience. The prompts prevent the AI from revealing its nature and maintaining its predefined role within the VR system.

The VR system uses OpenAI's API to integrate ChatGPT, handling various user inputs, including spelling and grammatical variations [73]. When input is unclear, responses may lack accuracy, necessitating validation of the NLP models. Whisper is used for voice transcription and translation, converting voice input into text for ChatGPT. The system also utilizes Whisper for voice transcription and translation into English so that users can interact using voice inputs, which are transcribed into text and passed to the ChatGPT module. Although a multilingual approach is not the focus of this study, the potential for using Whisper in future research remains viable [19,74]. Fig. 8 illustrates the Voice-to-Text-to-Speech process, converting voice input into text, processing it through ChatGPT, and generating a voice response. Additionally, AI trainer can recommend video-based training content based on user requests for further hazard information. Player data is dynamically managed through the 'PlayerInfo' C# script, ensuring tailored interactions within the training environment.

3.2.2. *AI tools integration for personalized safety training*

The iSafeTrainer integrates advanced AI tools, Text-to-Speech (TTS), Whisper, and ChatGPT, to create an interactive, human-centered training system that enhances engagement and accessibility by converting text into realistic audio, making it particularly valuable for educational and interactive applications [75]. Whisper, another AI tool by OpenAI, transcribes and translates voice input by tokenizing audio into a spectrogram and processing it through an encoder-decoder transformer, generating a text caption with special tokens for tasks like transcription and translation. [76]. When a user submits a voice input through the microphone, the whisper initiates a complex tokenization process, breaking down the audio into discrete units called tokens. The whisper architecture, implemented as an encoder-decoder transformer, converts the input audio into a spectrogram, which is then passed into an encoder. A decoder trained within the system predicts the corresponding text caption, enriched with special tokens that guide the single model to perform various tasks, including language identification, multilingual speech transcription, and speech translation to English or other languages.

First, Whisper is trained to detect speech absence and then assigns task-specific tokens for transcription or translation [74]. Its zero-shot performance across varied datasets and ability to minimize errors make it ideal for the transcription and translation functions of the system. After transcription, the text is routed to ChatGPT, which generates responses using advanced natural language processing. However,

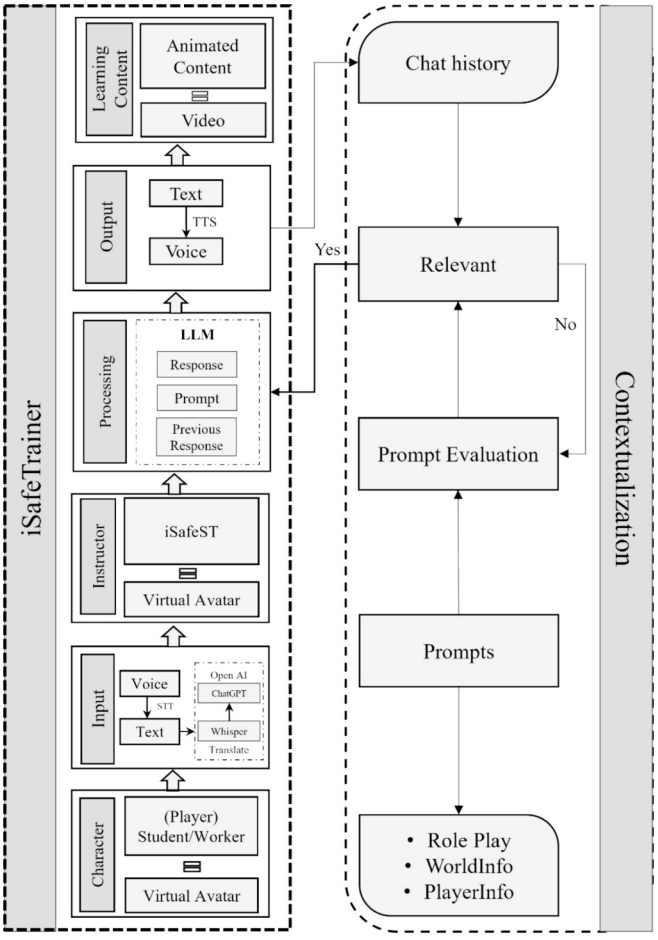


Fig. 8. Virtual Construction Sites **(a)** Electrical Safety Training Platform, **(b)** Scaffolding Safety Training Platform, and **(c)** Welding Safety Training Platform.

ChatGPT may produce irrelevant responses, so the VR system employs prompt engineering with contextualization techniques to refine answers. Contextualization incorporates details relevant to the construction domain, ensuring responses are tailored to participants' needs. Section 3.2.1 discusses how these prompts are crafted. Subsequently, the refined text-based response undergoes processing for text-to-speech conversion. The system integrates Whisper, ChatGPT, and TTS to

enable seamless data flow, facilitating accurate transcription, relevant response generation, and voice feedback. Fig. 9 presents a sequence diagram outlining API interactions, preprocessing, and data formatting, fostering effective communication, comprehension, and engagement in a more inclusive training environment.

3.2.3. Training Content Delivery

Using a constructivist learning theory approach, the AI NPC in iSafeTrainer not only enables users to engage with the Virtual environment but it can recommend video-based training content via clear directional guidance which helps users identify hazards detailed in section 3.1.1 and engage with scenarios that illustrate safe and unsafe construction practices.

A C# script controls video playback, allowing users to pause or play videos with specific commands (Shift + P for play, Shift + R for pause). The system is programmed to respond to specific queries related to training content, ensuring that users can access relevant material within the Unity environment. Fig. 10 provides an overview of the system's interactive functionalities and user interaction processes.

4. Case study and experiments

This section includes a case study example to provide users with a clear understanding of how to use the iSafeTrainer system. The example offers a brief overview of hazard interaction, mitigation strategies, and resource allocation within an experimental setup.

a. Learning Setup

Upon entering the virtual environment, participants are prompted to select their trade i.e., Electrical, Scaffolding, or Welding and their knowledge level: novice, beginner, or expert. They self-identified their trade and expertise level based on their academic background, coursework, or prior job-site experience. Students determined their category based on their involvement in relevant academic projects or coursework, while workers classified themselves according to their hands-on construction experience. This self-assessment ensured that the training content aligned with each participant's actual skill level and professional background. The iSafeTrainer system dynamically adapts training scenarios based on the participant's selection. For novices, the AI NPC provides step-by-step guidance, interactive prompts, and visual aids to enhance learning. For experienced participants, the system emphasizes advanced problem-solving tasks and hazard mitigation strategies, encouraging critical thinking and real-world application of safety concepts. This personalization enhances engagement and knowledge retention by ensuring that training remains both relevant and appropriately challenging.

For instance, a user selecting "Scaffolding Safety Training" at the beginner level would begin by pressing the "Start" button, entering a virtual environment populated with various hazards. These hazards are indicated by pop-up arrows, which serve to engage beginners and aid them in recognizing potential dangers. For learners unfamiliar with risk detection, this visual aid is crucial in fostering an intuitive learning experience. As a case in point, a beginner may only identify three out of four hazards present in the environment. Before asking about the hazards the AI trainer will ask the user "*Are you wearing appropriate protective equipment such as helmets, gloves, and safety boots?*" If user indicates non-compliance, the AI NPC provides corrective response and emphasizes the importance of PPE in reducing workplace hazards. After that AI NPC intervenes by asking whether all hazards have been found or not and when the user realizes one hazard remains unidentified, the AI trainer offers feedback and encourages them to search for the overlooked hazard. This active learning loop is vital for real-time engagement, allowing users to reflect on their hazard detection process. If a user struggles to locate all the hazards, AI NPC recommends supplementary video-based content to reinforce learning, it not only aids in safety hazard identification but also sharpens the user's mitigation skills. An illustrative system diagram highlighting the whole process is presented in Fig. 11.

b. Hazard Identification and Mitigation

In this phase, the user's interaction with specific hazards is detailed. For instance, Fig. 12(a) highlights how a user identifies the "missing planks" hazard, which poses a serious fall risk. In Fig. 12(b), the player observes a stack of planks nearby on the rooftop of the construction site and picks them up one by one to address the hazard. Finally, in Fig. 12 (c), the player successfully places the planks over the scaffold, eliminating the risk of falls and mitigating the danger. Through this process, users enhance their critical thinking and problem-solving abilities by actively engaging with the hazard within the Virtual environment which exemplifies how users can practically apply safety knowledge and collaborate with the system to resolve dangerous scenarios. By completing these tasks, users not only learn to identify hazards but also gain experience in real-world mitigation strategies, promoting a deeper understanding of workplace safety.

c. Experimental Setup

After developing the iSafeTrainer-based VR system, a structured experimental study was conducted to evaluate its feasibility and usability in tailored safety education. The experiment consisted of a single-

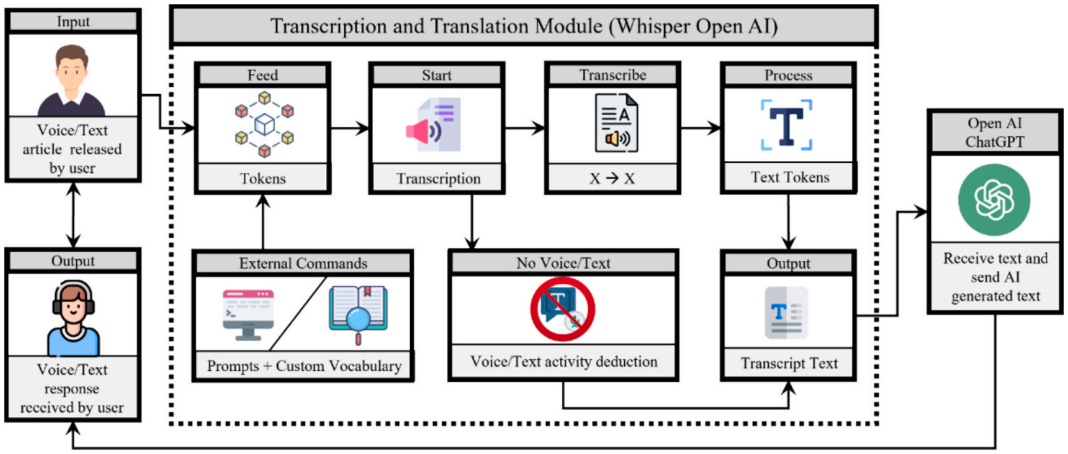


Fig. 9. Process Flow of Voice Input to Voice Output for Transcription and Translation of Interaction Between Tools



Fig. 10. **(a)** Learning Content in VR **(b)** C# Scripting to Control the Video Playback.

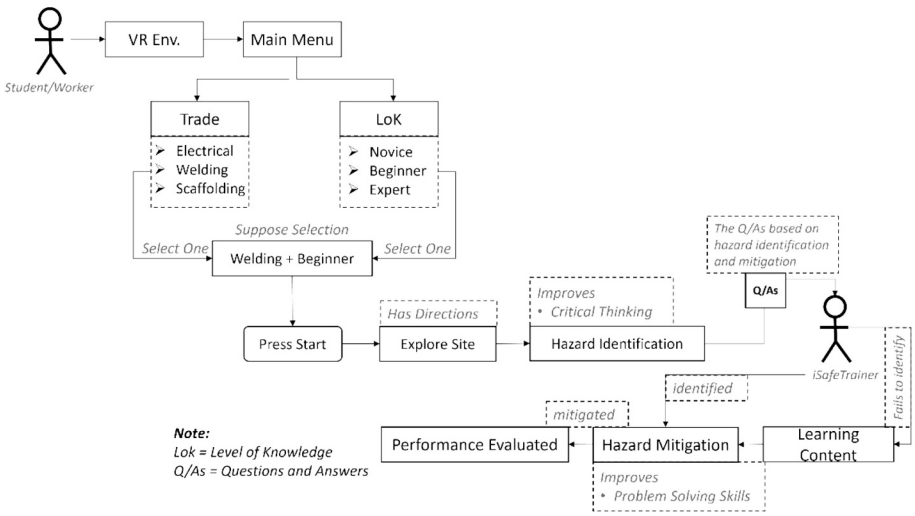


Fig. 11. System Diagram and Case Study for Interaction with iSafeTrainer for Personalized Safety Training

session evaluation for each participant, lasting approximately 30 min, divided into the following phases:

1. *Introduction (5 min):* Participants were briefed on the system's functionality, the role of AI NPCs, and the objectives of the study.
2. *Interaction with AI NPC (20 min):* Participants engaged with iSafeTrainer in a Virtual environment customized for their trade (e.g., electrical, scaffolding, welding). They interacted with the AI NPC through voice commands, enabling on-the-spot hazard identification and safety guidance.
3. *Survey Completion (5 min):* Participants completed a structured post-experiment questionnaire assessing usability, engagement, and perceived learning impact.

A total of 31 participants, including undergraduate and graduate students as well as post-doctoral candidates from Chung-Ang University, were meticulously recruited to represent a broad worker population. The participant pool consisted of 23 males and 8 females, covering a range of trades and knowledge levels. Most participants were construction or architectural students specializing in trades such as scaffolding or welding, indicating they were early on their educational journey, counting themselves as novices or beginners, post-doctoral researchers with prior job-site experience and expertise in specific trades. These individuals were actively engaged in their respective trades at the time of their study. This classification was self-reported. Detailed demographic information is summarized in Table 2. This study, conducted at Chung-Ang University, utilized a third-person desktop virtual environment and anonymized questionnaires completed by healthy adult volunteers (with non-identifiable demographic data (trade, age, knowledge level, and gender). According to institutional guidelines applicable at the time, formal Institutional Review Board (IRB) approval was not required.

4.1.1. Nature of interaction

Participants explored virtual hazardous construction sites using the iSafeTrainer VR system. Participants were free to navigate the virtual space, identifying and familiarizing themselves with various safety risks. After exploring the site, participants interacted with the AI-powered safety trainer who offers personalized safety guidance and recommendations based on the hazards encountered. All interactions with the AI NPC were conducted through voice-based queries, allowing participants to ask about hazards, safety measures, and best practices in real time. Examples of prompt include:

- Are you wearing appropriate protective equipment such as helmets, gloves, and safety boots?
- What are the hazards in this area?
- What precautions should I take while welding?
- How do I inspect scaffolding for safety compliance?

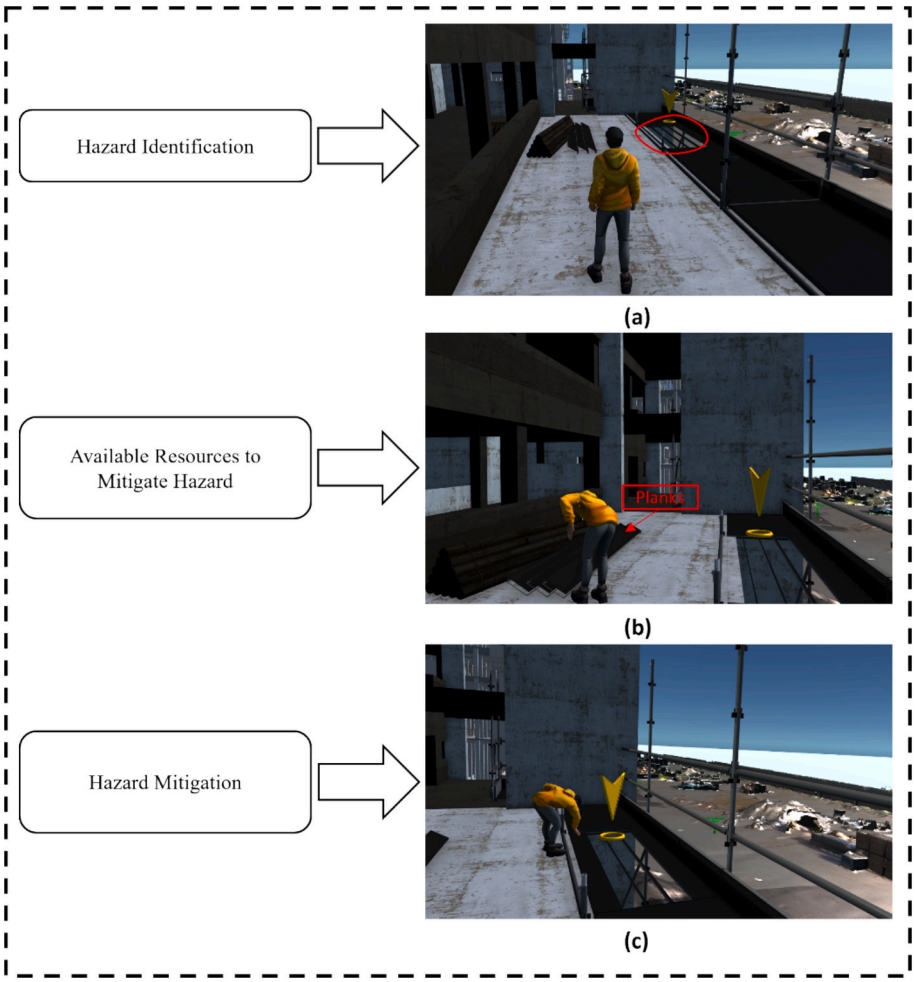


Fig. 12. **(a)**Identification of a Fall Hazard **(b)** Found Available Resources Around to Solve the Danger **(c)** Hazard Mitigation.

Table 2
Background information for participants.

Demographic Factors	Number (n = 31)	Percentage (%)
Gender		
Male	23	74 %
Female	08	25 %
Age		
18–20	05	16 %
25–30	19	61 %
30–39	07	22 %
Trade		
Electrical	14	45 %
Welding	09	29 %
Scaffolding	08	25 %
Knowledge Level		
Novice *(University students with coursework in a specific trade)*	15	48 %
Beginner *(post-doctoral researchers or students with limited hands-on experience)*	11	35 %
Expert *(post-doctoral researchers with prior job-site experience)*	05	16 %
Experiment Timeline		
Number of Sessions	1	
Session Duration (Minutes)	30	

The AI NPC provided trade-specific, actionable guidance, such as: "*Check the structural integrity of scaffolding before climbing and ensure guardrails are secured.*" Fig. 13 shows the participant communicating with AI NPC within iSafeTrainer in unity-based VR system for customized training.

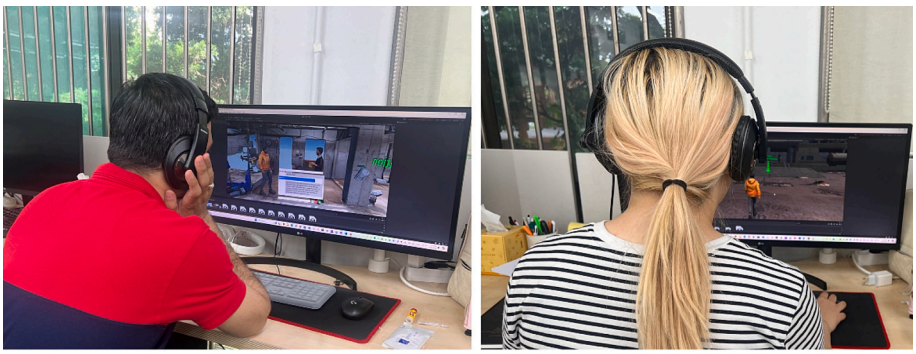
4.1.2. Survey metrics

To assess participant satisfaction and the effectiveness of the training, a multi-faceted survey based on validated instruments used in similar studies, such as user satisfaction scales [77] and system usability metrics [78] was employed, incorporating both structured and unstructured questions, designed to evaluate five key factors:

- User Satisfaction and Overall Usability
- User Experience and Interaction
- Training Effectiveness and Engagement
- System Guidance and Support
- Impact on Confidence and Preparedness

The structured questions utilized Likert scales and agreed/disagreed formats to gather quantitative data on various aspects, such as overall satisfaction, ease of interaction, perceived realism, usefulness of training content, and likelihood of recommendation. Validated instruments, including the System Usability Scale (SUS) [79,80] and the User Experience Questionnaire (UEQ) [81], were incorporated to provide standardized measures of usability and user experience.

Unstructured questions allowed participants to express thoughts and insights on topics such as voice interaction experience, personalized training, system guidance, and the impact on confidence when addressing real-world safety hazards. These provided qualitative data on the system's ability to foster critical thinking and problem-solving skills [20,82]. This experimental setup aimed to assess the feasibility and



(a) (b)

Fig. 13. (a) Participant With Expert Level of Knowledge Having Conversation with AI NPC On Desktop-Based Electrical Safety Training Platform **(b)** Participant with Beginner Level Of Knowledge Is Exploring The Scaffolding Safety Training Platform.

adaptability of the iSafeTrainer system rather than its long-term effectiveness. Future studies will incorporate detailed statistical validation and comparative analyses with traditional training methodologies to evaluate its impact on learning outcomes and hazard recognition skills.

5. Evaluation results

The iSafeTrainer system prioritizes personalized user experience and adaptive learning capabilities. While comparative analyses and advanced evaluation metrics were beyond the scope of this study, they are planned for future research. To assess the system's novelty, a qualitative comparison with recent studies was conducted, particularly examining their training methods and adopted technologies (Table 3). To further evaluate effectiveness of the developed system in tailored safety education, a questionnaire-based survey was conducted with construction students who participated in the VR program. The survey included both structured (questions 1–8) and unstructured (questions 9–15) items to collect quantitative and qualitative data (Appendix, (Table 6)). A detailed discussion of the experimental results is provided below, with mean results summarized in Appendix Table 7.

Table 3 presents a comparative analysis of recent studies in construction safety training, highlighting the unique attributes of iSafeTrainer, including personalized content delivery and the integration of conversational AI within a Virtual environment. These features set proposed systems apart from previous approaches and emphasize its contribution to advancing personalized training methodologies.

5.1. Key findings

User Satisfaction and Overall Usability: The results from questions 1, 5, 7, and 8 indicated a good level of overall user satisfaction with the iSafeTrainer system. 81 % of participants were satisfied with iSafeTrainer *(Q1)*, with 85 % noting improved safety knowledge and hazard mitigation skills *(Q7 & Q8)*.

User Experience and Interaction: Analysis of questions 2, 9, and 12 revealed insights into the user experience and interaction with the system. Most participants (85 %) found the voice interaction to be "easy" or "very easy" (Q2). However, the open-ended responses in Q9 and Q12 highlighted some areas for improvement and additional features including animations like, talk, walk, or head movement for realistic human nature of AI trainers.

Training Effectiveness and Engagement: The survey results indicated that the tailored safety education provided by iSafeTrainer was effective and engaging. 79 % of participants "Neutral", "agreed" or "strongly agreed" that the system provided personalized training tailored to their trade and knowledge level (Q3), and 83 % found the virtual environment realistic and immersive (Q4). Additionally, the open-ended responses in Q11 and Q13 revealed that participants found communication for real-time guidance and feedback to be particularly engaging and beneficial aspects of the personalized training.

System Guidance and Support: 78 % to 80 % Participants provided examples of safety hazards encountered in the virtual environment and described how the AI NPC within iSafeTrainer guided them in mitigating these hazards (Q10). Common themes included how to place missing guardrails for scaffolding, which material is more dangerous in case of welding etc. Furthermore, in Q14, participants described situations where the system's guidance was particularly helpful, such as Users asking AI NPC for relevant training content on overhead power lines received video-based guidance. These responses suggest that the system provides effective guidance and support in real-world scenarios.

Impact on Confidence and Preparedness: The survey results indicated a positive impact of the system on users' confidence and preparedness in addressing safety hazards on construction sites. 75 % of participants reported being "Neutral", "likely" or "very likely" to recommend the system to other construction workers (Q6), suggesting a good level of confidence in the system's effectiveness.

The open-ended responses in Q15 revealed that the system, realistic, collaborative, and personalized nature helped the novice and beginner players in terms of improving their confidence and preparedness for real-world jobsite situations. In addition to the previously provided overall means of questionnaire-based surveys and the results based on the five factors discussed, group surveys were also evaluated based on the participants' level of knowledge. Table 4 presents the evaluation of five key factors related to user satisfaction and training effectiveness across three user groups: Novice, Beginner, and Expert. Novices consistently rated the system the highest, with scores above 80 %. So as

Table 3
Comparison of different studies with the attributes of the proposed system.

Year	Research Study	Training Method			Technology Adopted		Experimental Results		Ref.
		One-size	Personalized	Content Delivery	VR	LLMs	Group	Survey	
2023	Li et al.	x	o	x	o	x	o	x	[14]
2023	Wen et al.	o	x	x	o	o	x	o	[38]
2024	Rane et al.	o	x	x	x	o	x	o	[83]
2024	Hussain et al.	o	x	o	o	o	o	x	[19]
2024	Proposed Study	x	o	o	o	o	o	o	

Table 4
Results of All Five Key Factors Corresponding to The Level Of Knowledge Of All 31 Participants.

Key Factors	Novice	Beginner	Expert
KF1: User Satisfaction and Overall Usability	86.3%	82.3%	15.0%
KF2: User Experience and Interaction	87.6 %	77.6 %	14.7 %
KF3: Training Effectiveness and Engagement	87.7 %	80.5 %	16.0 %
KF4: System Guidance and Support	86.0 %	76.4 %	14.0 %
KF5: Impact on Confidence and Preparedness	89.3 %	69.1 %	8.0 %

the beginners rated from 77 to 82 % but Experts were more critical, with ratings between 8 % and 16 %, indicating a need for system refinement for experienced users.

Table 5 summarizes the survey results across user groups. Satisfaction rates were calculated from Question 1 (Q1), where 81 % of all participants reported satisfaction, with values for beginners, novices, and experienced users drawn directly from Table 4 (KF1: User Satisfaction). Engagement scores combine the ease of voice interaction (Q2 mean: 4.3/5) and qualitative feedback (Q9, Q12), with subgroup scores derived from Table 4 (KF2: User Experience) percentages converted to a 5-point scale. Safety knowledge gains reflect responses to Q7 and Q8, where 85 % of all users reported improvement, and subgroup values align with Table 4 (KF3: Training Effectiveness). From Table 5, "*Safety knowledge gain reflects participants' self-reported improvement in understanding hazards and mitigation skills (Q7 & Q8). Engagement scores combine ease of interaction (Q2) and qualitative feedback (Q9, Q12), 86.3% of novices reported high satisfaction, compared to 15% of experienced users. Perceived improvement in safety knowledge exceeded 80% for beginner users, highlighting the system's effectiveness for early-career learners*".

6. Discussion, limitations and future work

Discussion: The integration of VR and conversational AI (LLMs) advances construction safety training methodology. This approach addresses existing limitations through adaptive training that responds to individual trade expertise and knowledge levels to enhance their learning outcomes compared to traditional one-size-fits-all methods. The system leverages advanced AI technologies, including Whisper for speech recognition and ChatGPT for interactive dialogue, creating an immersive and impactful learning experience. Evaluation results showed high levels of user satisfaction, particularly among novices, aligning with findings by Admi et al. (2023) that demonstrated improved performance in VR-based training compared to traditional methods. [84]. Experienced users suggested more advanced scenarios to enhance training depth. It is important to note that the system prototype was validated using unsafe scenarios specific to building projects. However, the system's modular architecture allows for adaptation to various construction contexts, such as bridges, highways, or industrial facilities. The Virtual environment can be customized to include relevant hazards, and the AI NPC can be updated through prompts to expand its knowledge and applicability to other types of projects.

Limitations: Despite the system's contributions to personalized safety training, several limitations warrant discussion. System performance depends significantly on VR and AI technological capabilities. Challenges such as hardware performance, software stability, and the accuracy of natural language processing have the potential to hinder the overall user experience and training outcomes. Virtual environment uses a third-person perspective controller to avoid motion sickness but may limit the immersive experience compared to fully immersive systems like head-mounted displays. Evaluation methods in this study were primarily survey-based and focused on usability and satisfaction, with no quantitative metrics like task completion rates or retention scores included. These metrics are planned for future research to provide a more rigorous validation of the system's effectiveness. Additionally, external factors such as environmental conditions, participants' prior experience, and organizational safety culture were not accounted for, which may influence training outcomes. The participant pool consisted primarily of individuals with similar educational backgrounds in construction and architecture and from a single geographic region, potentially limiting the generalizability of the findings to broader populations or industries. Finally, while LLMs can be trained on extensive datasets, there may still be limitations in terms of their depth and breadth of knowledge in the specific domain of construction safety. Ensuring comprehensive and up-to-date safety expertise across all relevant domains may require extensive training and curation of the language model. Effective safety training often requires empathy, encouragement, and the ability to gauge the emotional state of the user. Integrating these soft skills into the conversational abilities of iSafeTrainer may be challenging, as current LLMs still have limitations in emotional intelligence and social awareness.

Table 5
Summary of Post-Training Survey Results.

Metric	Overall (N = 31)	Novice Users (N = 15)	Beginner Users (N = 11)	Experienced Users (N = 05)
Satisfaction Rate	81 %	86.3%	82.3%	15.0%
Engagement Score	4.3/5	4.38/5	3.88/5	0.74/5
Perceived Improvement in Safety Knowledge	85 %	87.7 %	80.5%	16.0%

Future Work: Currently, this study focuses on the initial development and feasibility testing of the iSafeTrainer system. While descriptive analysis of survey responses provides valuable insights, future studies will incorporate comparative and statistical analyses and additional evaluation metrics such as task completion rates, hazard recognition scores, and learning retention with advanced visualization to rigorously validate the system's effectiveness in diverse scenarios and across different user groups. Future enhancements will focus on developing LLM-integrated virtual instructors capable of demonstrating safety tasks. These intelligent agents can simulate real-world safety scenarios at the request of the trainees, offering interactive and informative demonstrations to enhance trainees' comprehension and retention of safety protocols. This integration not only represents a significant step towards enhancing safety training initiatives in the construction, empowering trainees with practical knowledge and skills essential for on-site safety but also opens new possibilities for the advancement of safety training methodologies, ultimately contributing to improved safety standards within the construction sector. Likewise, enabling the LLM-integrated instructor to autonomously recommend video-based safety training content from a web database could significantly enhance the customization and effectiveness of safety programs which align well with the trend towards automated, AI-driven solutions in occupational safety and health training, ensuring programs remain up to date with the latest safety standards and practices.

7. Conclusion

This paper introduced iSafeTrainer, a personalized construction safety training system integrating LLMs with a third-person perspective in a non-immersive desktop virtual environment. This system was designed to address the heightened workplace risks faced by construction workers, providing personalized, interactive, and trade-specific safety education. The evaluation of iSafeTrainer demonstrated its feasibility and user satisfaction in delivering personalized safety training. The system's ability to dynamically adapt to individual expertise and trade requirements showcased its potential to overcome the limitations of traditional, static training methods. The results from questionnaire-based surveys provided valuable insights into usability and user satisfaction.

The implications of the study extend beyond construction safety,

advancing personalized digital learning methodologies. By integrating LLMs, such as ChatGPT, within an interactive third-person perspective in a non-immersive desktop VR framework, iSafeTrainer offers a learner-centered training model that adapts dynamically to experience levels of users, trade-specific challenges, and learning preferences. A key contribution of this system is its ability to provide real-time, AI-driven safety instruction while also recommending relevant video-based training materials, creating a multi-modal learning experience that reinforces knowledge retention. This approach ensures that safety training is not only interactive but also tailored to the diverse trades, expertise levels, and learning styles of construction workers and students.

While this study provides valuable insights, limitations include the third-person perspective controller, while avoiding motion sickness, may limit the immersive experience compared to fully immersive systems. Evaluation methods were primarily survey-based, focusing on usability and satisfaction without quantitative metrics like task completion rates or retention scores. Future research should focus on incorporating comparative and statistical analyses, along with additional evaluation metrics such as task completion rates, hazard recognition scores, and learning retention with advanced visualization, to rigorously validate the system's effectiveness in diverse scenarios and across different user groups. Future enhancements will focus on developing LLM-integrated virtual instructors capable of demonstrating safety tasks, offering interactive and informative demonstrations to enhance trainees' comprehension and retention of safety protocols.

CRediT authorship contribution statement

Aqsa Sabir: Writing – original draft, Supervision, Software, Project administration, Methodology, Investigation, Data curation, Conceptualization, Formal analysis, Resources, Validation, Visualization. **Rahat Hussain:** Resources, Formal analysis, Conceptualization. **Akeem Pedro:** Writing – review & editing, Data curation, Conceptualization. **Chansik Park:** Supervision, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT to improve English writing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Table 6
Full questionnaire.

Questionnaire-based Survey
Structured Questions:
1. Rate your overall satisfaction with the iSafeTrainer system.
(1 = Very Dissatisfied, 2 = Dissatisfied, 3 = Neutral, 4 = Satisfied, 5 = Very Satisfied).
2. How easy was it to interact with the iSafeTrainer using voice commands?
(1 = Very Difficult, 2 = Difficult, 4 = Normal, 4 = Easy, 5 = Very Easy)
3. The iSafeTrainer provided personalized safety training tailored to my trade and knowledge level.
(1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)
4. I found the virtual reality environment realistic and immersive.
(1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)
5. Rate the usefulness of the video-based training content recommended by the iSafeTrainer.
(1 = Not Useful at All, 2 = Not Useful, 3 = Normal, 5 = Useful, 5 = Extremely Useful)
6. How likely are you to recommend the iSafeTrainer system to other construction workers?
(1 = Not at All Likely, 2 = Not Likely, 3 = Neutral, 4 = Likely 5, = Extremely Likely)
7. The iSafeTrainer system improved my understanding of safety hazards and mitigation strategies in terms of personalized safety training.
(1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)
8. Overall, how would you rate the usability of the iSafeTrainer system?
(1 = Very Poor, 2 = Poor, 3 = Normal, 4 = Good, 5 = Excellent)
Unstructured Questions:
9. Describe your experience interacting with iSafeTrainer. What worked well, and what could be improved?
10. Can you provide an example of a safety hazard you encountered in the virtual environment and how the iSafeTrainer guided you in mitigating it?
11. What did you find most engaging or beneficial about the personalized safety training provided by the iSafeTrainer?
12. If you could suggest any additional improvements to the iSafeTrainer system, what would they be?
13. How did you experience the iSafeTrainer system compared to traditional safety training methods?
14. Describe a situation where the iSafeTrainer's response or guidance was particularly helpful.
15. How did the iSafeTrainer system impact your confidence and preparedness in addressing safety hazards on construction sites?

Table 7
Overall questionnaire survey-based results.

Questions	Mean	Results
1. Rate your overall satisfaction with the iSafeTrainer system on a scale of 1 to 5.	4.1	Satisfied Score illustrates that users felt at ease and satisfied when using the virtual screen for learning.
2. How easy was it to interact with the iSafeTrainer using voice commands?	4.3	Easy Score shows it was easy to interact with iSafeTrainer using voice commands but needs improvement
3. The iSafeTrainer provided personalized safety training tailored to my trade and knowledge level.	4.1	Agree iSafeTrainer provided most relevant and adequate answers.
4. I found the virtual reality environment realistic and immersive.	4.1	Agree The score describes those participants who mostly agreed on the realistic nature of Virtual environment.
5. Rate the usefulness of the video-based training content recommended by the iSafeTrainer.	4.0	Normal It seems that users can easily familiarize themselves with using the keyboard, mouse, and headphones to operate the system.
6. How likely are you to recommend the iSafeTrainer system to other construction workers?	3.7	Likely It seems that proposed system can be recommended to others.
7. The iSafeTrainer system improved my understanding of safety hazards and mitigation strategies in terms of personalized safety training.	4.0	Agree The personalized learning, communication and interaction of the system help users to pay attention and remember tailored safety knowledge.
8. How would you rate the usability of the iSafeTrainer system?	4.3	Good Seems like iSafeTrainer system is useful according to the personalized training requirements.
9. Describe your experience of interacting with the iSafeTrainer. What worked well, and what could be improved?	4.0	Good Interacting with intelligent safety trainer was helpful and most of the communication seems effective.
10. Can you provide an example of a safety hazard you encountered in the virtual environment and how the iSafeTrainer guided you in mitigating it?	4.0	Likely Most of the players provided examples of hazards they faced in the Virtual environment and tried to mitigate them.
11. What did you find most engaging or beneficial about the personalized safety training provided by the iSafeTrainer?	4.2	Useful Answers show the tailored communication between iSafeTrainer, and player was more beneficial
12. If you could suggest any additional improvements to the iSafeTrainer system, what would they be?	4.0	Helpful Improvements related to iSafeTrainer's animations like, talk, walk, or head movement was provided
13. How did you experience the iSafeTrainer system compared to traditional safety training methods?	4.0	Very good Very good scores highlight that users find the proposed more effective for personalized training as compared to traditional methods
14. Describe a situation where the iSafeTrainer's response or guidance was particularly helpful.	3.7	Clear A clear situation has been described in which the user got the effective guidance from iSafeTrainer.
15. How did the iSafeTrainer system impact your confidence and preparedness in addressing safety hazards on construction sites?	3.9	Satisfied Results show that this system made the players enough capable to deal with real -world hazards effectively.

Data availability

Data will be made available on request.

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